

DPS

DIAGNOSTIC PREDICTION SUITE

Machine Learning Powered Risk Screening for
Diabetes & Cardiovascular Disease

Ravada Siddharth · Arpit Kumar · Kanak Sharma

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Minor Project · Academic Year 2026-27

UNDER THE SUPERVISION OF DR. KUSUM LATA MAM
DEPT. OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE



Dual Disease Module

Diabetes + Cardiovascular



Gradient Boosting

90.25% Accuracy / AUC 0.956



Random Forest

82.25% Accuracy / AUC 0.901



Django Web App

AJAX · SQLite · Chart.js



SMOTE Balanced

2,000 Samples per Model

Python 3.10+

Django 5.x

scikit-learn

Chart.js

SQLite

THE GLOBAL BURDEN OF PREVENTABLE CHRONIC DISEASE

537M+

Adults with diabetes
globally (IDF 2025)

~50%

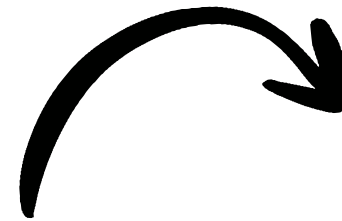
Cases undiagnosed
until complications arise

17.9M

Deaths from CVD
per year (WHO)

THE PROBLEM

- Lab tests (HbA1c, fasting glucose) are costly
- ~50% diabetic cases missed by conventional screening
- Rural India lacks accessible diagnostic infrastructure
- CVD symptoms often unnoticed until severe



OUR SOLUTION

- Instant, low-cost prediagnostic web screening
- Dual-disease coverage (Diabetes + CVD) in one tool
 - 90.25% accuracy · AUC 0.956 via ML + SMOTE
- Early intervention reduces complications by 58%

From single-model script → full-stack dual-disease web app

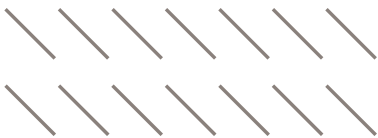
v1–v2 · OLD VERSION

- Single disease: Diabetes only
- Random Forest — 768 samples, no SMOTE
- 8 raw features, no engineering
- No web interface — Jupyter notebook
- No result persistence or database
- No visualisation charts
- No PDF export
- Class-imbalanced training data
- Accuracy ~76–80% | AUC ~0.84



v3.1 · DPS (CURRENT)

- Dual disease: Diabetes + Cardiovascular
- GradientBoosting + RandomForest, 2000 samples
- 11 features (3 engineered: GlucoseAge, BMIAge, InsulinGlucose)
- Full Django 5.x web application (AJAX)
- SQLite database — all predictions persisted
- Radar · Feature Importance · Distribution charts
- Branded PDF export via jsPDF
- SMOTE balanced (1000 : 1000 per class)
- Accuracy 90.25% / 82.25% | AUC 0.956 / 0.901

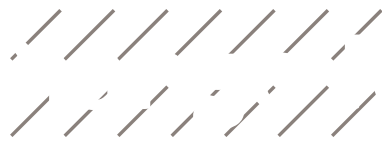


TWO TRAINED CLASSIFIERS — TWO DISEASE MODULES

 DIABETES MODULE	
Algorithm	Gradient Boosting Classifier
Dataset	Pima Indians — UCI ML Repo
Samples	2,000 (SMOTE — 1k : 1k)
Features	8 raw + 3 engineered = 11
Engineered	GlucoseAge · BMIAge · InsulinGlucose
n_estimators	250 max_depth: 5 lr: 0.08
Test Accuracy	90.25%
ROC-AUC	0.9557
Top Feature	GlucoseAge — 32.34% importance

 HEART DISEASE MODULE	
Algorithm	Random Forest Classifier
Dataset	Synthetic — Framingham/Cleveland
Samples	2,000 (SMOTE+undersampling)
Features	13 clinical parameters
Key Features	Age · CP type · Thalach · Oldpeak · CA
n_estimators	300 max_depth: 12 balanced
Test Accuracy	82.25%
ROC-AUC	0.9007
Top Feature	ca (major vessels) — highest importance

SYSTEM ARCHITECTURE



Presentation Layer

index.html · Chart.js · jsPDF · Web Audio API · CSS Day/Night mode



Application Layer

views.py · forms.py · urls.py · AJAX JsonResponse · Django ORM



ML Service Layer

ml_service.py · GradientBoosting · RandomForest · StandardScaler

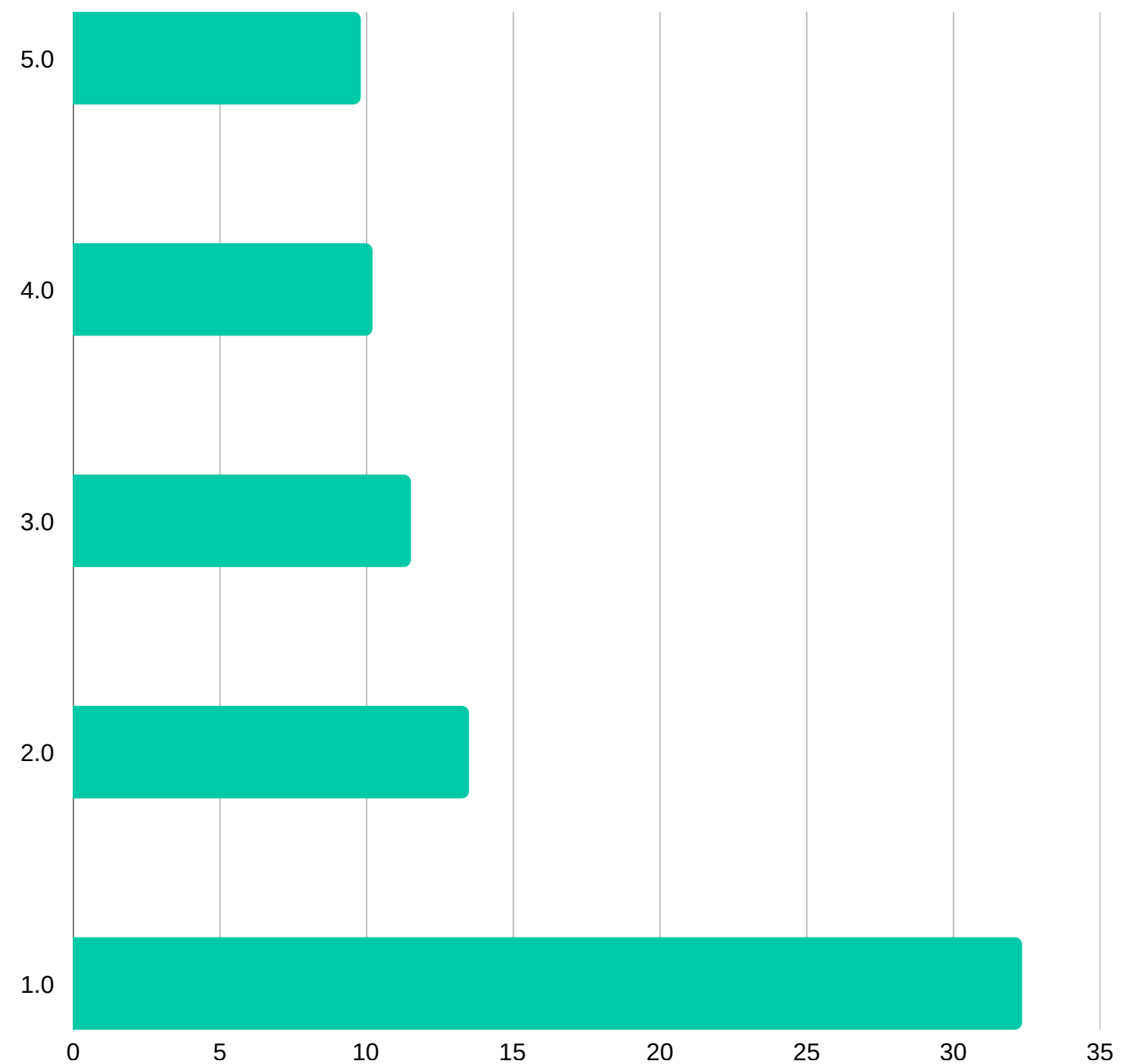
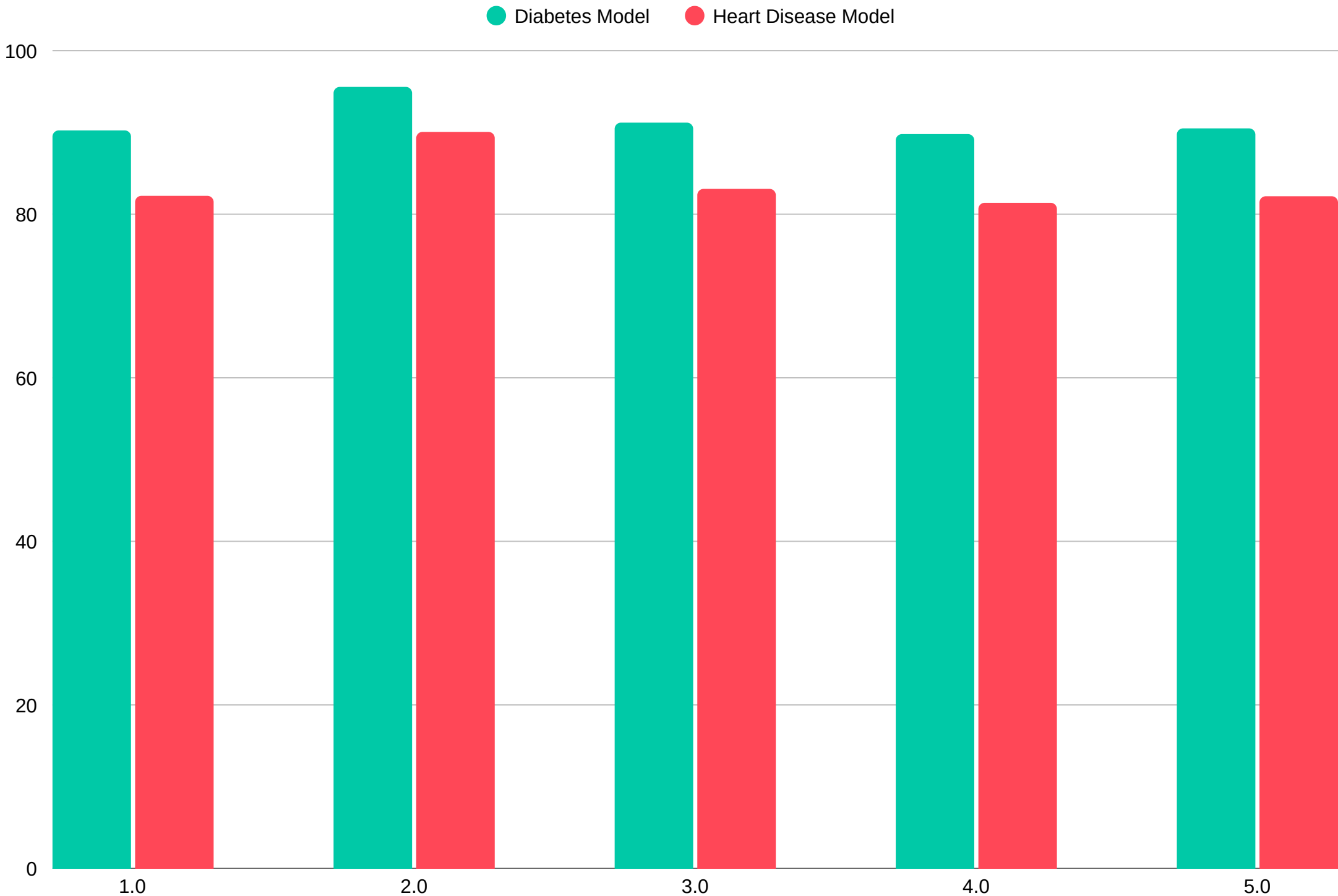


Data Layer

SQLite3 · PredictionRecord model · JSONField · Auto-timestamp



Accuracy, ROC-AUC & Feature Importance



90.25%
Diabetes Accuracy

0.9557
Diabetes ROC-AUC

82.25%
CVD Accuracy

0.9007
CVD ROC-AUC

* Engineered features outperform raw inputs in feature importance ranking

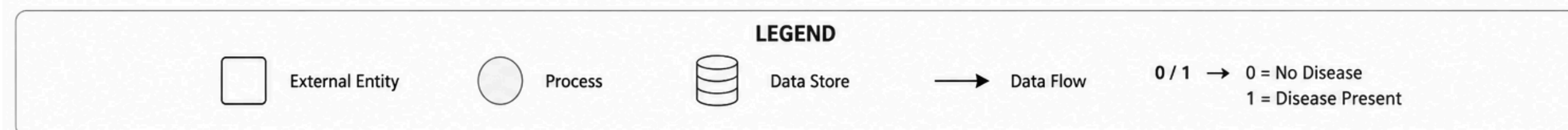
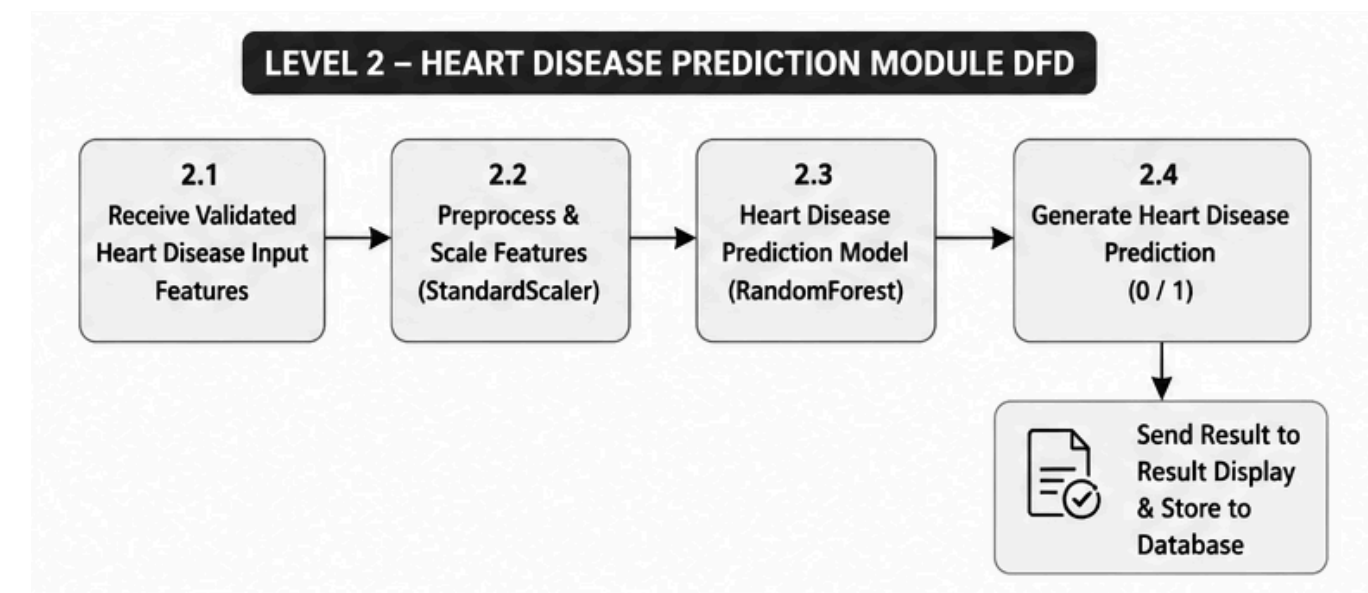
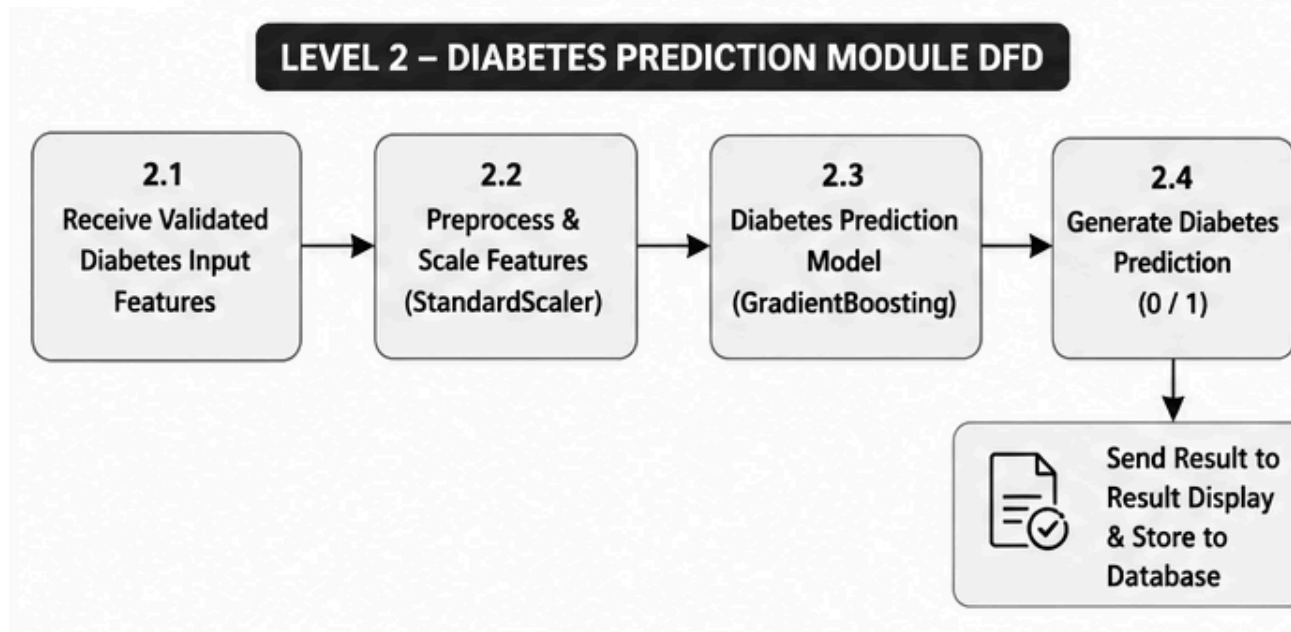
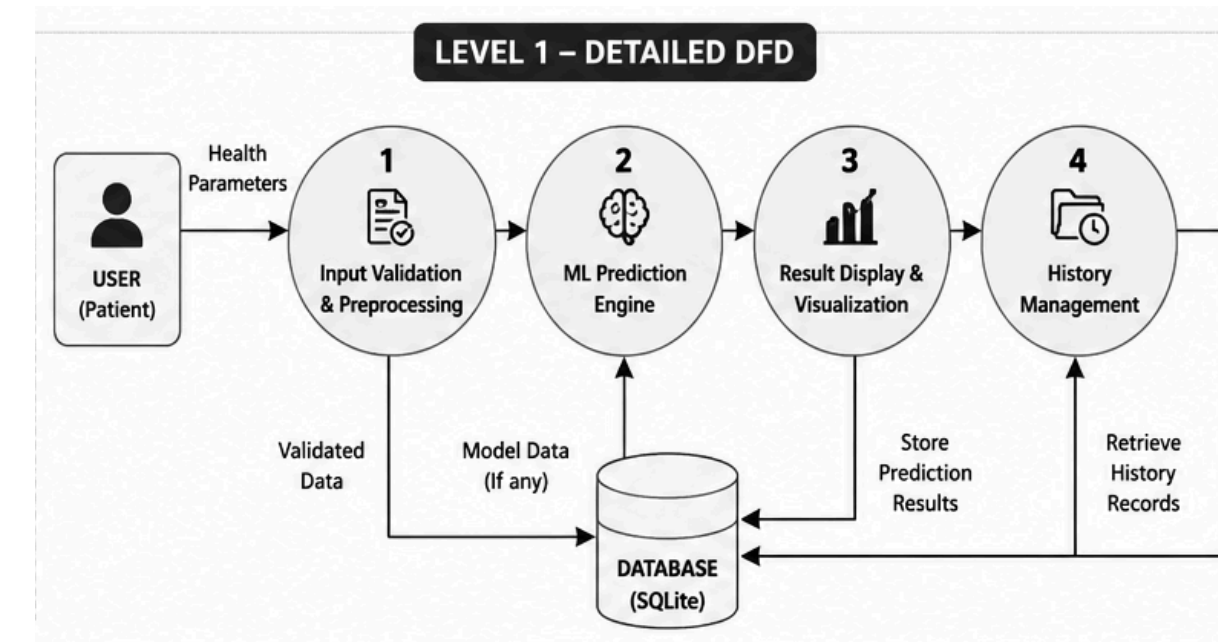
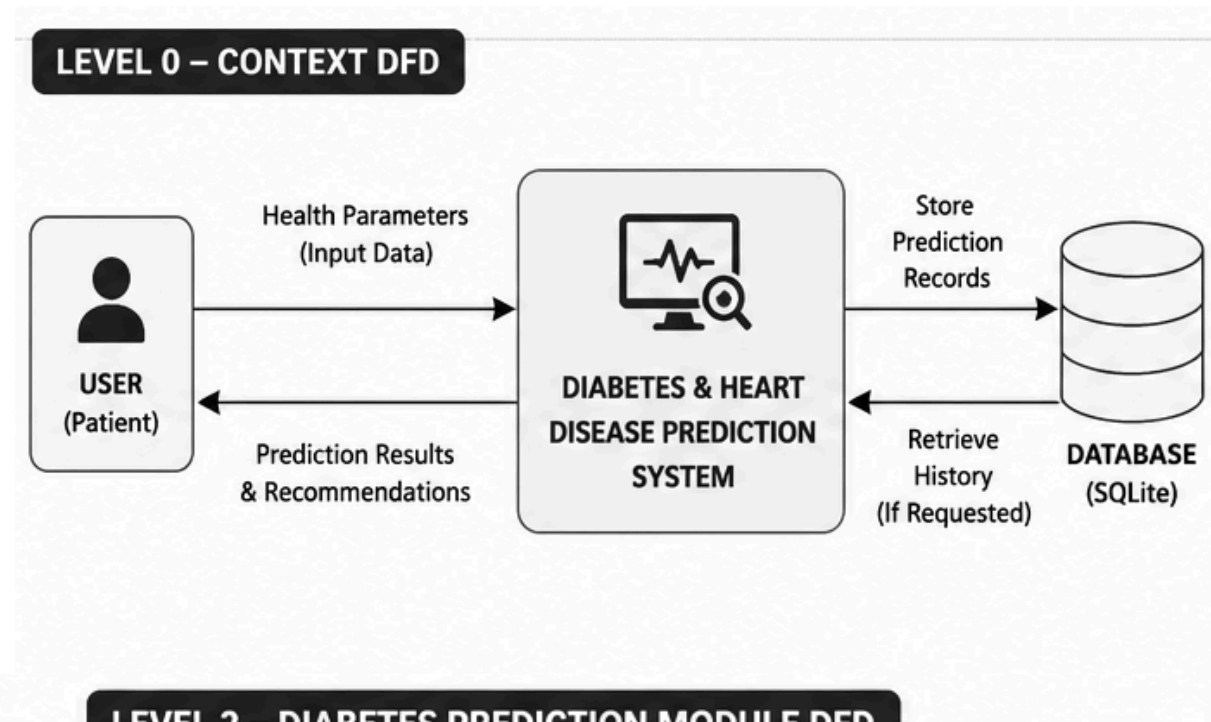
METHODOLOGY PIPELINE

End-to-end flow: data → trained model → live prediction



USER-SYSTEM DATA FLOW

Layer-wise structure of the proposed ML-based diagnostic system



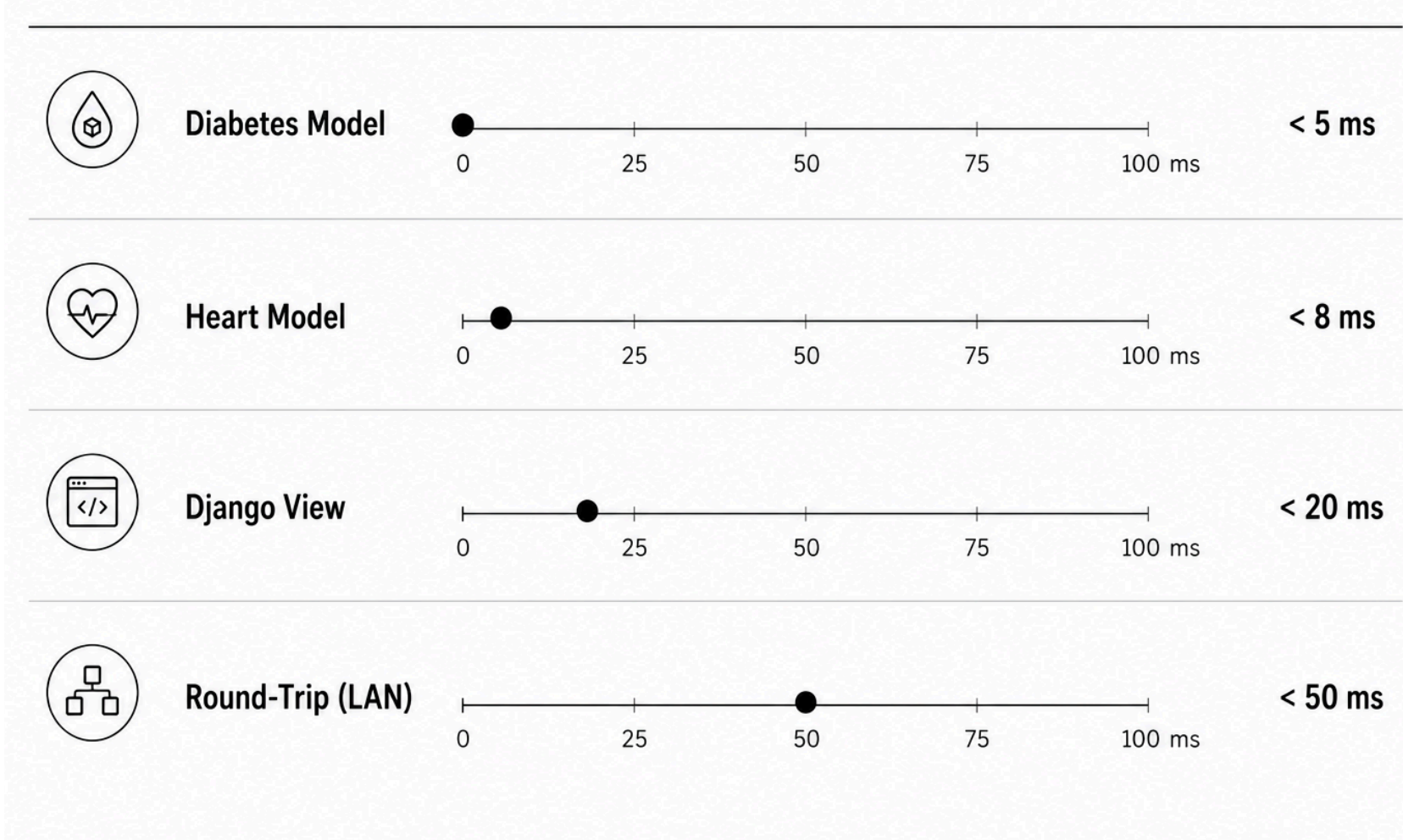
Test cases, accuracy metrics & functional verification

Sample Functional Test Cases

Test Case	Input	Result	Pass?
Low-risk diabetes	Glucose=85, BMI=22, Age=25	< 30%, Low	✓
High-risk diabetes	Glucose=200, BMI=40, Age=55	> 60%, High	✓
Zero imputation	Glucose=0 (missing value)	Replaced ~121	✓
Invalid input	Age = -5	400 error JSON	✓
Low-risk CVD	Age=35, chol=160, CP=0	< 30%, Low	✓
High-risk CVD	Age=70, exang=1, ca=3	> 60%, High	✓
PDF export	Click Export after prediction	A4 PDF saved	✓
History persistence	9 sequential predictions	Last 8 shown	✓

Avg. inference latency: Diabetes < 5 ms · Heart < 8 ms · Full round-trip < 50 ms (LAN)

Performance Testing



Thank You

Questions & Discussion

Ravada Siddharth

ML Integration &
Data Pre-Processing

Arpit Kumar

ML Engineer
& Data Analyst

Kanak Sharma

Frontend dev

- ◆ 90.25% diabetes accuracy · AUC 0.9557
- ◆ 82.25% CVD accuracy · AUC 0.9007
- ◆ Full-stack Django web app — AJAX, SQLite, Chart.js, jsPDF
- ◆ SMOTE-balanced 2,000-sample datasets per model

